

- 1 (a) $GM/R^2 = R\omega^2$ C1
 $\omega = 2\pi / (24 \times 3600)$ C1
 $6.67 \times 10^{-11} \times 6.0 \times 10^{24} = R^3 \times \omega^2$
 $R^3 = 7.57 \times 10^{22}$ M1
 $R = 4.23 \times 10^7 \text{ m}$ A0 [3]
- (b)(i) $\Delta\Phi = GM/R_e - GM/R_o$ C1
 $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) (1 / 6.4 \times 10^6 - 1 / 4.2 \times 10^7)$
 $= 5.31 \times 10^7 \text{ J kg}^{-1}$ C1
 $\Delta E_p = 5.31 \times 10^7 \times 650$ C1
 $= 3.45 \times 10^{10} \text{ J}$ A1 [4]
- (c) e.g. satellite will already have some speed in the correct direction ... B1 [1]